

Test Documentation: Tier 1 - Core VTXO Verification

Overview

This suite focuses on the structural and cryptographic validation of the VTXO DAG, ensuring that every virtual transaction is authentic and correctly anchored to the Bitcoin blockchain.

Test Goals

1. **DAG Reconstruction:** Programmatically building the ancestor-child relationship from the **VTXO Root** down to the commitment anchor.
2. **Signature Integrity:** Verifying that every transaction in the DAG was signed by the appropriate MuSig2/Schnorr aggregated keys (including internal and tweaked keys).
3. **On-chain Anchor:** Confirming the **Anchoring Leaves** of the virtual tree are correctly anchored to a confirmed commitment transaction.

Environment

- **Providers:** `MockIndexerProvider` and `MockOnchainProvider`.
- **Data Source:** Simulated Regtest/Signet transaction structures (PSBTs and on-chain status).
- **Execution:** Node.js/Vitest environment.

Evaluated Scenarios

Scenario	Description	Expected
Valid DAG	A consistent 3-level Reversed-DAG with correct signatures and confirmed commitment.	PASS
Tampered Sig	A single bit change in the Schnorr signature of an ancestor node.	FAIL (INVALID_SIGNATURE)
Missing Link	A DAG where a child input doesn't correctly reference its ancestor's outpoint.	FAIL (INPUT_CHAIN_BREAK)
Ghost Commitment	A valid DAG whose anchoring commitment is not found on-chain.	FAIL (COMMITMENT_NOT_FOUND)
Anti-Mirage Attack	ASP provides a fake TXID or malformed hex during anchoring.	FAIL (ORACLE_POISONING_DETECTED)
RPC Spoofing	Bitcoin RPC returns HTTP 200 with result: null or malformed JSON.	FAIL (INVALID_RPC_RESPONSE)

Results

- **Success Rate:** 100% of functional and security tests passed (87/87 total tests).
- **Latency:** Sub-50ms for typical 5-depth DAG reconstruction and validation.
- **Security:** Successfully rejected all 20+ simulated “Mirage”, “Incorrect Signature”, and “RPC Poisoning” attacks.